



CSE461

Introduction to Robotics Lab

Real-World Implication

Group : 3

Section : 2

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“Autonomous Noise and Air Quality Monitoring Robot”

1. Introduction:

The task of our project is to construct a mobile robotic system that will be capable of measuring the level of pollution and investigating the environment on its own. Air and sound pollution have become more common and significant environmental hazards, particularly in the environs of overcrowded towns, manufacturing zones, schools, and workplaces. The traditional manual monitoring is limited in coverage, poor, and uneven. The robot is inexpensive with autonomous and continuous monitoring.

The synthesis of an MQ135 gas sensor is controlled by an Arduino Uno R3 in the process of controlling the robot. sound sensor component, and an ultrasonic sensor to detect obstacles. The robot provides dynamically collected information in place of fixed point measurement with fusion and real-time perception movement. These automated systems can play a significant role, contrasted with early warning of danger, education, and judgment in the dynamism of the increasing need for sustainable environmental practices.

2. Purpose:

The central aim of the task is to develop a robot capable of capturing the environment, information, making self-movement, and notifying the users at an acceptable time when the values collected are satisfactory. The robot also has the MQ135 sensor to detect the amount of gas in the air, such as CO₂, NH₃, and smoke, and the sound sensor to identify the amount of noise. The robot may have a pattern of zigzag or grid-scan movement to travel to multiple places with increased levels of efficiency and a higher precision of environmental mapping. Within the daily setting, such as schools, factories, and suburbs, such an initiative aims to provide a feasible and viable method of monitoring the levels of pollution. Such information would be used by governmental bodies or other environmental agencies to identify the location of pollution and take proactive measures to prevent pollution. The robot can also be demonstrated as an instance of embedded technologies, robotics, sensor implementation, and environmentally friendly technologies.

3. Societal implications:

By increasing the collection and utilization of environmental data, this robotic system has the ability to significantly help society. Daily observation assists in detecting pollution spikes in residential areas, businesses, schools, and institutions. Communities can make accurate choices because of consistent mobility (like grid-based monitoring), which guarantees accurate information from many different locations. By immediately recognizing hazardous locations, LED/buzzer notifications make the

system more readily available to regular people. Long-term information can assist local governments in promoting healthier living conditions, improving the development of cities, and upholding laws. This affordable mobile robot provides a beneficial and significant answer for populations without the use of heavy monitoring equipment.

4. Health implications:

Anxiety, difficulties with hearing, breathing issues, and permanent health hazards are all closely connected to air and noise pollution. A validated MQ135 sensor is used by the robot to detect the dangerous level of gas more accurately, which will give accurate health-related data.

Similarly, converting sound sensor data into the form of safe, moderate, and dangerous decibels provides rapid and practical data about noise exposure.

The robot can identify the unsafe areas, a school with an unhealthy environment, or a company with excessively noisy machinery, since it can move across various areas and not repair itself in a single location. An early warning will help the local people and the authorities to ensure that they take proactive measures ahead of health issues.

5. Safety implications:

Continuous monitoring of the environment supports the increase in safety standards in areas like manufacturing facilities, workplaces, and classrooms. Once the pollution is at dangerous levels, the robot notifies the users of the situation through LEDs or buzzers. Its automatic obstacle-preventing system provides assurance of the safety of movement around people and objects and prevents accidents. The sensor is disconnected from sound vibrations in the motor, which causes false readings, and only the alarms are given out when there is real danger, so as to avoid false alarms due to noise in the machine. This robot minimizes blind spots, and the way it reviews the surroundings is cautious, making environmental monitoring safer and more reliable.

6. Legal implications:

Governmental agencies set the proper standards of noise and air quality.

uphold environmental regulations. In a number of countries, including Bangladesh, there are legal limits imposed on pollution in open places, in schools, and in business establishments.

Companies can ensure that they are not behind the curve with the laws concerning the environment through the information that this robot provides. As a case, though learning institutions are mandated to offer safe learning.

Businesses, which are faced with the environment, are obligated to watch over pollution.

Disobedience may lead to legal consequences, penalties, or restrictions in the areas of operation.

The robot is applicable as an instrument for collecting evidence by keeping historical records of the level of pollution in the past. The issues about privacy should also be considered. The robot does not go against the rules on privacy since it gathers environmental information as opposed to collecting personal information. Nonetheless, on placement, it should be done with care so that the robot does not interfere with essential operations or restricted areas.

7. Cultural implications:

The cultural tendency of many communities is to reduce the safety of the environment, unless there is a severe concern. This robot raises awareness about the surroundings by rendering pollution evident and understandable with the help of real-time data and alarms. The robot can facilitate the learning of experience and motivate students to come up with sustainable practices in learning institutions. The availability of transparent data on the environment by societies could help them to formulate a culture of sustainability and duty of care. This encourages the change of society in the long-term to favor the more sustainable and healthier lifestyles.

8. Conclusion:

The Autonomous Noise and Air Quality Monitoring Robot is a simple, inexpensive, and economical device to maintain a watch over the environment. It handles severe issues like air pollution, noise risks, and demands of the robotic surveillance system with its navigational autonomy, pollution detection, and real-time warning options. The practical effects of it include gains in health, social awareness, safety at workplaces, legal compliance, and cultural changes, among others. As the issues of population and the environment grow, such robotic technologies can become significant to developing healthier, safer, and smarter communities.